AMMOS:
Automated Massbank MS2 toolkit to Obtain Spectras

User Manual (Jupyter Interactive)

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Abstract

This manual describes an interactive Jupyter tool designed to obtain MS2 data spectra from MassBank, parse peak lists and annotations, and export standardized `.dat` files together with metadata logs (`README`). This document explains the workflow, the role of each user control, how the tool handles data sources (API versus MassBank data archive), and common troubleshooting scenarios. Examples are included to illustrate typical use patterns without exposing implementation code.

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1 What the program does

The toolkit acts as a guided workflow for converting MassBank MS2 records into useful MassSpectra data. Conceptually, it performs five tasks:

1. **Record discovery (optional):** Query MassBank servers for MS2 records matching a compound name and obtain a list of candidate accessions.
2. **Record loading:** open the official *MassBank data* archive (a ZIP containing canonical record text files) either from a local ZIP or by downloading the GitHub ZIP into memory.
3. **Peak parsing:** extract peak information (m/z and relative intensity) from the `PK$PEAK` section of each record.
4.
5. **Formula matching (optional):** if the record provides `PK$ANNOTATION`, map formulas to peaks using a ppm tolerance.
6. **Export:** generate one `.dat` file per accession and append a structured Markdown entry into an individual molecule `README` log (instrument, ion mode, collision energy, resolution, etc.).

1.1 Inputs and outputs

Inputs

- A **molecule/compound name** (for searching) *and/or* a list of **MassBank accessions** (manual paste).
- A **ppm tolerance** controlling how strictly formulas are assigned to peaks.
- A **MassBank data ZIP source**: either a local ZIP file path or the GitHub ZIP.

Outputs

- **One `.dat` file per accession** with three columns:
  (a) `intensity` (as the relative abundance concept, in MassBank typically 0–1000),
  (b) `mz` (m/z),
  (c) `formula` (empty if no match).
- **One `README.md` per molecule** containing a chronological log of exported accessions and their metadata for each molecule. It is a kind of notebook or labbook

2 User interface overview (Jupyter widgets)

The tool is designed for Jupyter Notebook/Lab and presents an interactive panel. The following controls appear:

Molecule
Text field used to enter the compound name for API-based discovery of MS2 records.

Search MS2
Button that triggers the API query and populates the record selection list.

MS2 list
A multi-selection list showing candidate records as *accession — title*. You can select one or more.

Accessions (optional)
Text area where you may paste accessions manually, one per line. This is used if you prefer not to rely on API search or if the API is temporarily unavailable.

ppm tol
Numeric control setting the maximum ppm error allowed to match an annotated formula to a measured peak.

Local ZIP
Optional path to a local `MassBank data` ZIP. If left empty, the tool downloads the GitHub ZIP in memory.

Load selected
Button that loads records from the ZIP, parses spectra, and shows a preview.

Export .dat + README
Button that writes output files into the output directory.

Output panel
Notebook area where summaries, previews, and export paths are printed.

3 Workflow: step-by-step explanation

3.1 Step 1 — Choose how you want to identify spectra

You can identify the spectra to load in two ways:

Option A. Search by compound name (recommended for exploration): Enter the compound name in the *Molecule* field and press *Search MS2*. The program will query one or more MassBank API endpoints and list candidate MS2 records.

Option B. Paste accessions manually (recommended for reproducibility): If you already know the record IDs (e.g., from a paper, MassBank web interface, or prior runs), paste them (one per line) into the *Accessions* box. In this mode you do not need the API step.

3.2 Step 2 — Provide the MassBank data archive source

The program loads the canonical record texts from the *MassBank data* archive, which exists as a ZIP.

- **Local ZIP:** If you have already downloaded the archive, set *Local ZIP* to your path. This avoids repeated downloads and is faster.
- **GitHub ZIP:** If the *Local ZIP* field is empty, the program downloads the GitHub `main.zip` into memory and reads it without creating a ZIP file on disk.

3.3 Step 3 — Load and parse selected records

Press *Load selected*. For each accession, the program:

1. Locates the corresponding `<ACCESSION>.txt` file inside the ZIP.
2. Extracts peak lines from the `PK$PEAK` section.

3. Ensures that intensities are expressed as *relative percent*. If only absolute intensities are present, it rescales them so that the maximum becomes 100.
4. Extracts annotation lines from `PK$ANNOTATION` if present.
5. Extracts metadata fields (instrument, ion mode, collision energy, etc.) from the record header.
6. Builds an internal table (DataFrame) with `intensity`, `mz`, `formula`, and `accession`.
7. Make an interactive plot which ones represent the data frame loaded

At the end of this step, the tool prints a short summary table (per accession) and a preview of the first lines of the combined peak table.

3.4 Formula attachment (ppm tolerance)

If `PK$ANNOTATION` is present, the record may provide fragment formulas associated with annotated m/z values. The tool assigns formulas to measured peaks by:

- computing the ppm error between each peak m/z and each annotated m/z,
- choosing the closest annotation,
- accepting the match only if the error is $\leq$ the user-defined **ppm tol**.

If no annotation is available or no annotation falls within tolerance, the formula field for that peak remains empty.

3.5 Step 4 — Export files

Press *Export .dat + README*. The tool creates the output directory and then for each accession:

- writes a `.dat` file containing the three columns (`intensity`, `mz`, `formula`),
- appends a Markdown section for each molecule `README.md` log:
  - record title,
  - accession,
  - timestamp of export,
  - instrument and method metadata,
  - number of peaks and number of peaks with matched formulas.

To avoid duplication, the README append operation checks whether the accession already exists in the README content. You are free to customize this part according to your needs.

4 Pipeline graph

Figure 2 summarizes the end-to-end logic.

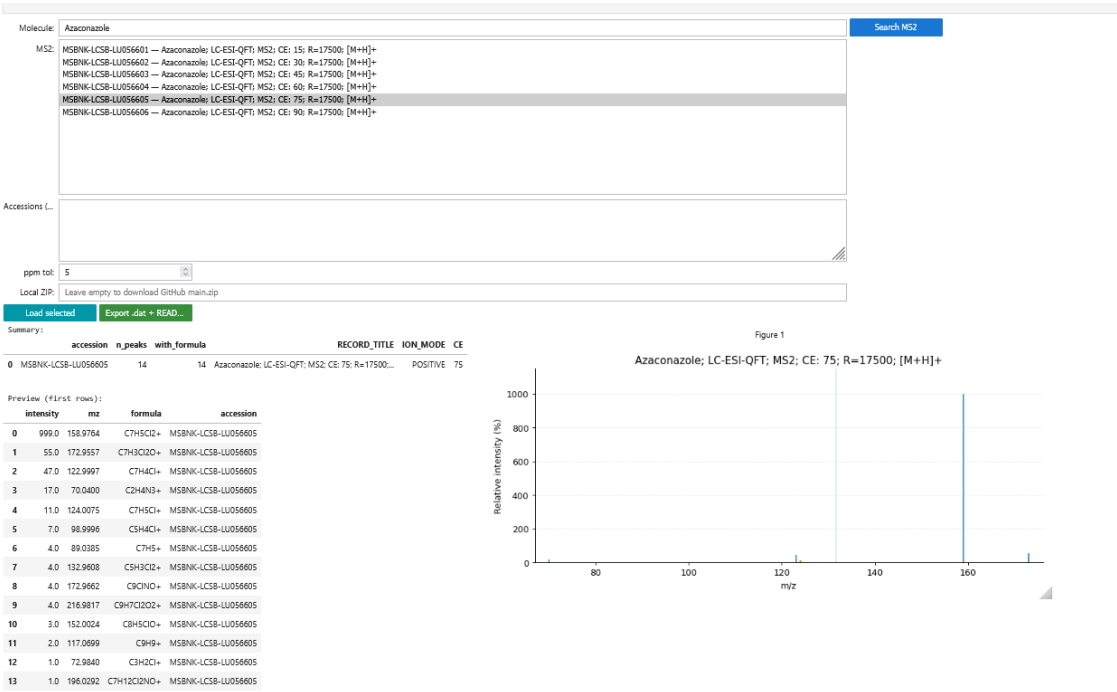

Figure 1: Graphical interface of AMMOS.

5 Usage examples

5.1 Example 1 — Exploratory search by compound name

Goal: quickly locate MS2 spectra available for a compound and export a subset for analysis.

Step 1. In *Molecule*, type `Azaconazole` (or any compound name you expect MassBank to recognize).

Step 2. Click *Search MS2*.

Step 3. In the *MS2 list*, select the accessions you want to analyze (e.g., select multiple collision energies or instruments).

Step 4. Keep *Local ZIP* empty if you do not have the archive locally.

Step 5. Set *ppm tol* to a value appropriate for your instrument class (typical values: 2–10 ppm for high-resolution data).

Step 6. Click *Load selected* and inspect the preview.

Step 7. Click *Export .dat + README*.

Step 8. Check the output directory (`massbank_dat`) for:

- `.dat` files (one per accession),
- `<Molecule>_README.md` with metadata entries.

5.2 Example 2 — Reproducible run using manual accessions

Goal: reproduce exactly the same dataset across machines or at a later date.

Step 1. Leave *Molecule* empty (optional).

Step 2. Paste a curated list of accessions (one per line) into *Accessions (optional)*.

Step 3. Provide a *Local ZIP* path if you want stable behavior without network dependency.

Step 4. Choose *ppm tol* and keep it fixed for all related exports.

Step 5. Click *Load selected* and verify the summary counts.

Step 6. Click *Export .dat + README*.

5.3 Example 3 — Choosing an appropriate ppm tolerance

Goal: avoid incorrect formula assignments.

General guidance:

- **High-resolution instruments** (Orbitrap/QTOF): start with 2–5 ppm.
- **Moderate resolution**: consider 5–10 ppm.
- If you observe many peaks receiving incorrect formulas, **decrease** ppm tolerance.
- If you observe almost no formulas being assigned despite annotations, **increase** ppm tolerance slightly and confirm instrument calibration.

6 Interpretation of outputs

6.1 `.dat` files

Each `.dat` file is designed to be easy to import into analysis workflows. The columns have the following meaning:

`intensity`
Relative intensity (%), typically scaled so that the most intense peak is 1000.

`mz`
m/z value of the peak.

`formula`
Fragment formula assigned via annotation matching; empty if unknown or unmatched.

Peaks are sorted primarily by descending intensity and secondarily by ascending m/z, making it easy to inspect top fragments.

6.2 Per-molecule `README.md`

The README is a running log that supports reproducibility. Each entry typically includes:

- accession and record title,
- date/time when the export was generated,
- instrument name/type,
- MS type, ion mode, collision energy, and resolution when available,
- number of peaks and number of formula matches.

7 Data sources and what is downloaded

7.1 API calls

When using name search, the tool sends small HTTP requests to MassBank API endpoints and receives JSON responses. These responses are not stored as files.

7.2 MassBank data ZIP

If *Local ZIP* is empty, the tool downloads the MassBank data GitHub ZIP into memory and reads it directly. It does not automatically save that ZIP to disk. If you provide a local ZIP, no download is required.

7.3 Files created locally

Local files are created only during the *Export* step: `.dat` files and `README.md` logs in the configured output directory.

8 Common issues and troubleshooting

8.1 Python version and typing

If your environment uses older Python versions, certain may fail. The acript includes a minimum version guard (Python $\geq$ 3.8). If your Jupyter kernel is older, upgrade the kernel or run the notebook in a newer conda environment.

8.2 API returns no results

Possible reasons:

- Temporary network or server outage.
- The compound name does not match MassBank naming.
- The record exists but is not labeled as MS2 in the API filters.

Solution: paste accessions manually or search MassBank via the web interface to obtain accessions.

8.3 Accession not found in ZIP

The accession may not exist in the ZIP version you are using, or the accession may be mistyped. Check:

- spelling and capitalization,
- whether your local ZIP is up to date,
- whether the accession corresponds to MassBank (not another database).

8.4 No PK$PEAK section

Some records may be incomplete or use unexpected formatting. The tool requires a valid `PK$PEAK` block to export a spectrum.

8.5 Formulas not assigned

This can happen if:

- `PK$ANNOTATION` is missing in the record,
- the ppm tolerance is too strict,
- annotated m/z values do not match peaks within tolerance.

9 Recommended best practices

- For reproducibility, keep a local copy of the MassBank data ZIP and reference it via *Local ZIP* if you want an offline mode.
- Export datasets by accession lists (manual mode) for stable reruns across machines and dates.
- Keep ppm tolerance consistent within a study and document it in your methods section.
- Use the README logs as a guide for which spectra were used.

10 Contact and citation

If you use this toolkit in academic work, please cite:

FrustratedAlchemist20. (2026). *FrustratedAlchemist20/AMMOS_JL_Toolkit: AMMOS JL Toolkit v1.0 (v1.0)*. Zenodo. https://doi.org/10.5281/zenodo.18882841

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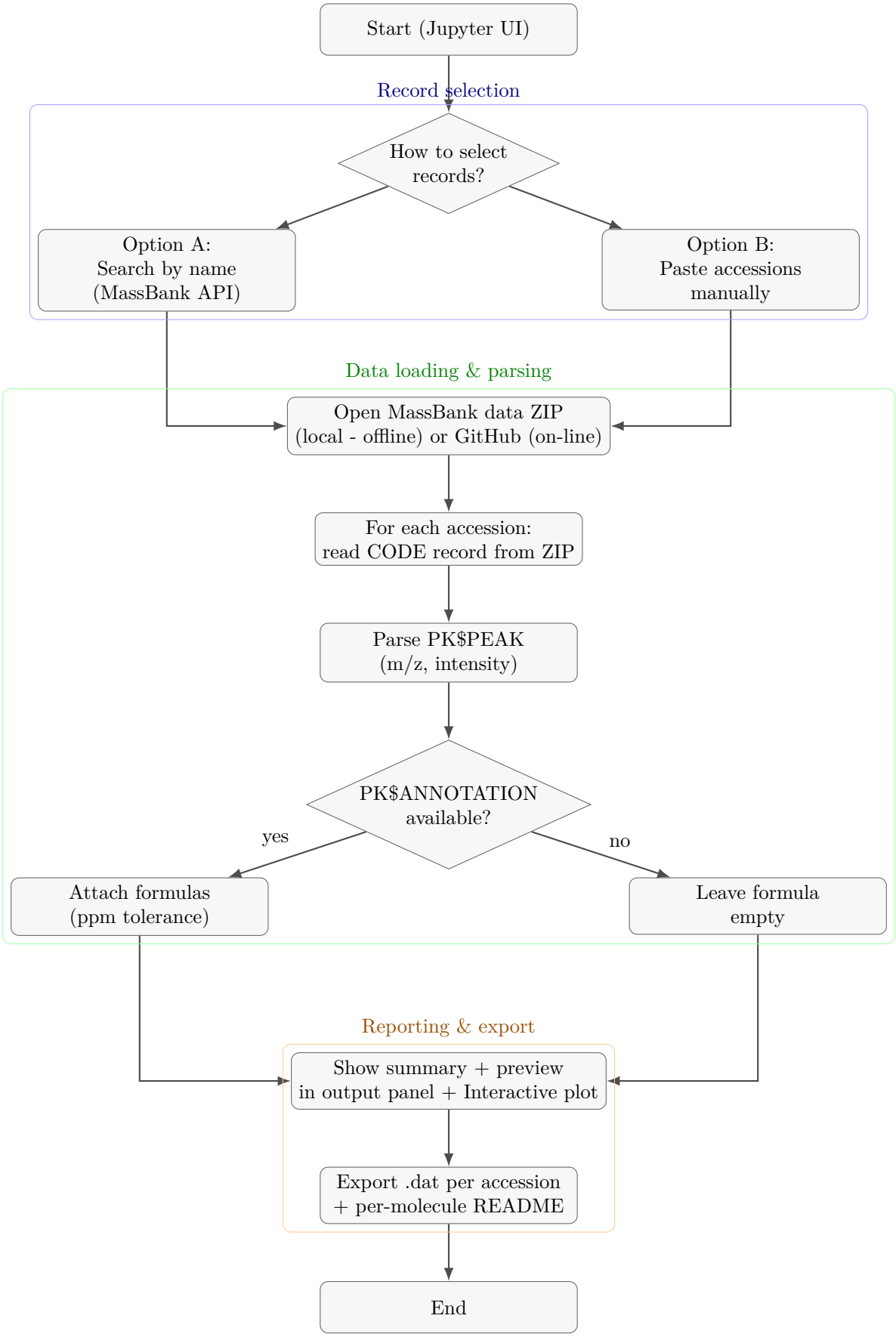

Figure 2: High-level workflow of the MassBank MS2 Spectrum Extraction Toolkit.